04-SEP-2022

Histograms

→

Ages = { 4, 12, 14, 13, 24, 26, 30, 35, 36, 37, 40, 41, 42, 43, 50, 51, 65, 68, 78, 90, 95, 100 }

Steps to create histogram:

1. Sort the numbers
2. Bins → No. of groups
3. Bins size = $\frac{max - min}{bins}$

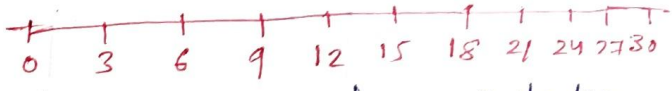
e.g: [0, 10, 20, 25, 30, 35, 40]

min = 10
max = 40
bins = 10
bins size = $\frac{40-10}{10} = 3$

→ If data starts from 0,
bins size = $\frac{max}{bins}$

→ If data starts from other values except 0,
bins size = $\frac{max - min}{bins}$.

→ Wrong



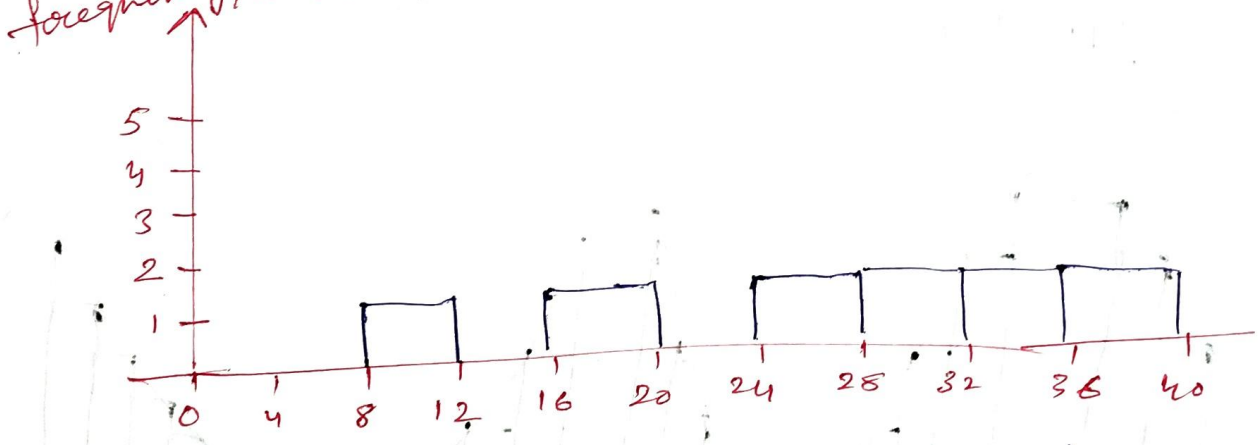
value ranging from 0 to 40 but here it is only till 30.

Continuous variable

correct

e.g: [0, 10, 20, 25, 30, 35, 40], bins = 10

bins size = $\frac{40}{10} = 4$



→ We can decide bins size based on our choice.

e.g: Ages = { 10, 12, 14, 18, 24, 26, 30, 35, 36, 37, 40, 41, 42, 43, 50, 51, 65, 68, 78, 90, 95, 100 }

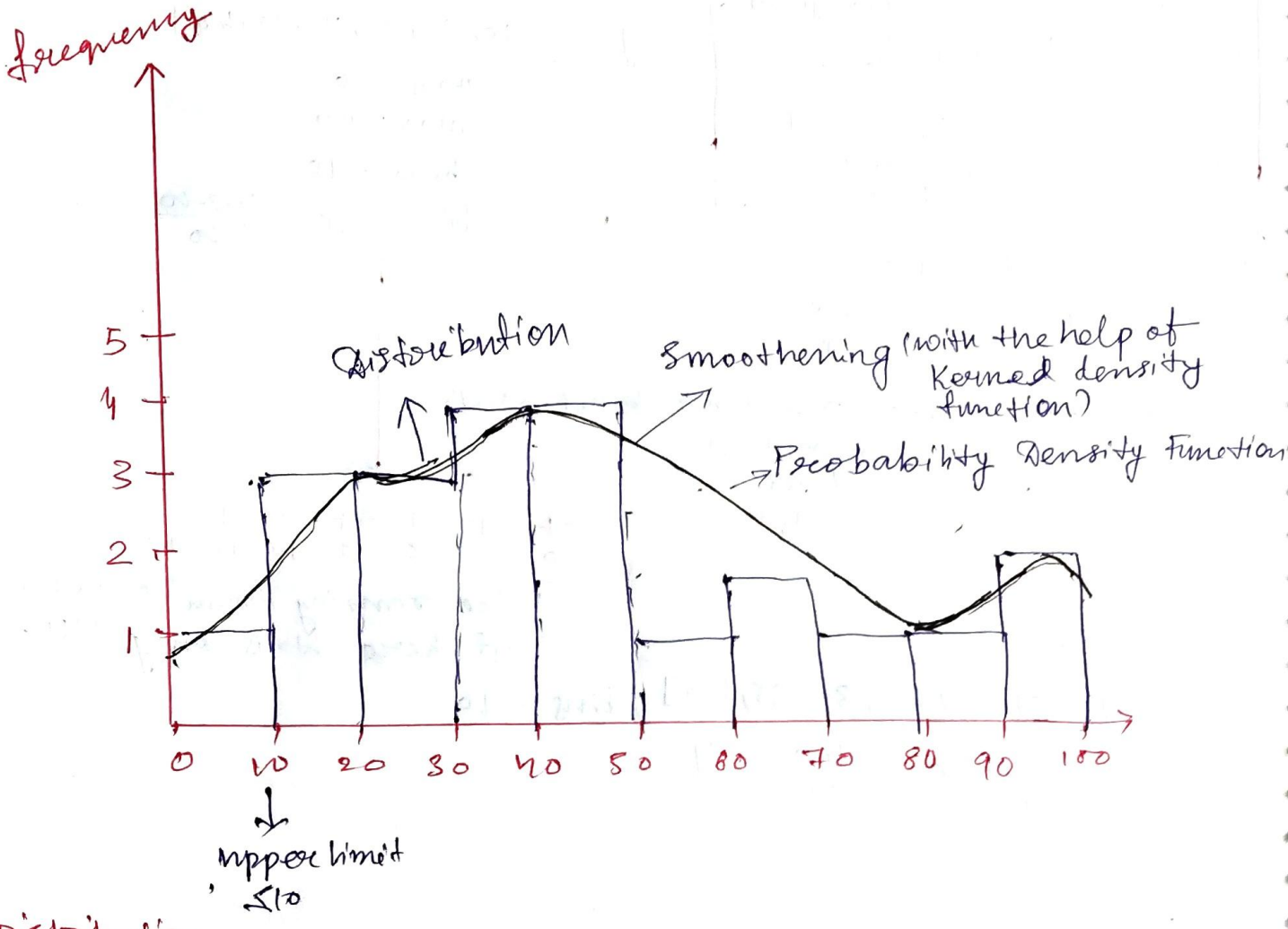
bins = 10

bins size = $\frac{100}{10}$ = 10

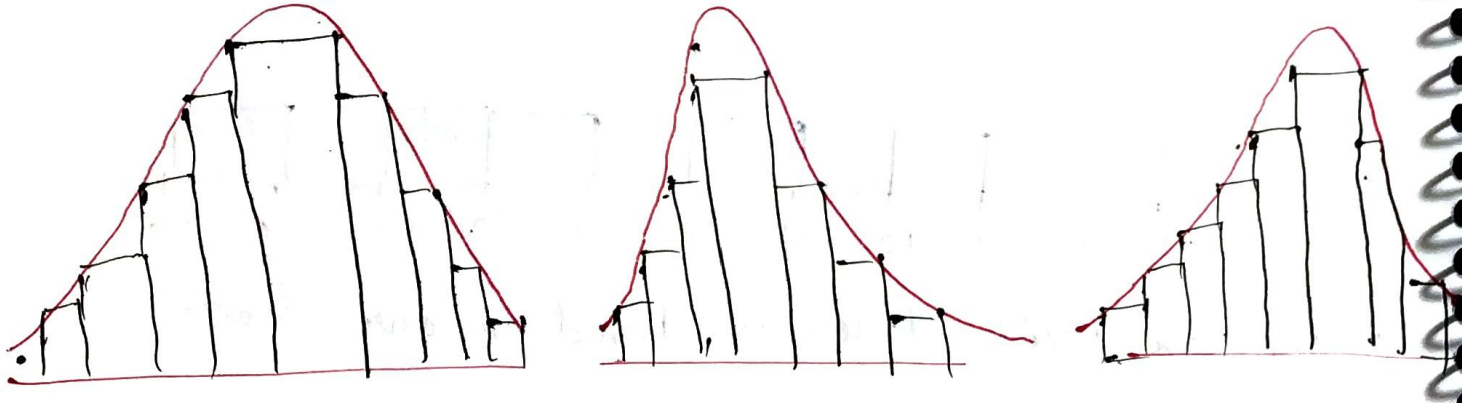
our choice

bins = 20

bins size = $\frac{100}{20}$ = 5



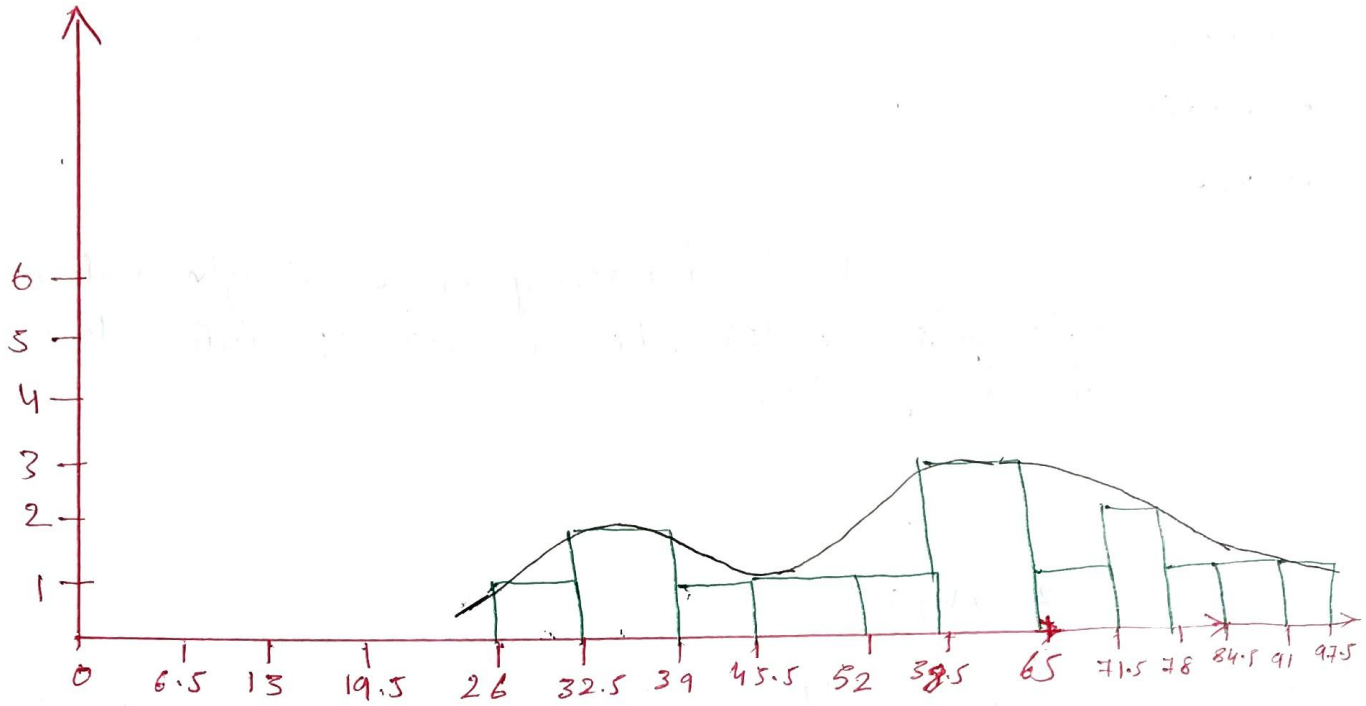
Distribution



e.g: Weight = {30, 35, 38, 42, 46, 58, 59, 62, 63, 68, 75, 77, 80, 90, 95}

Let's take bins = 10

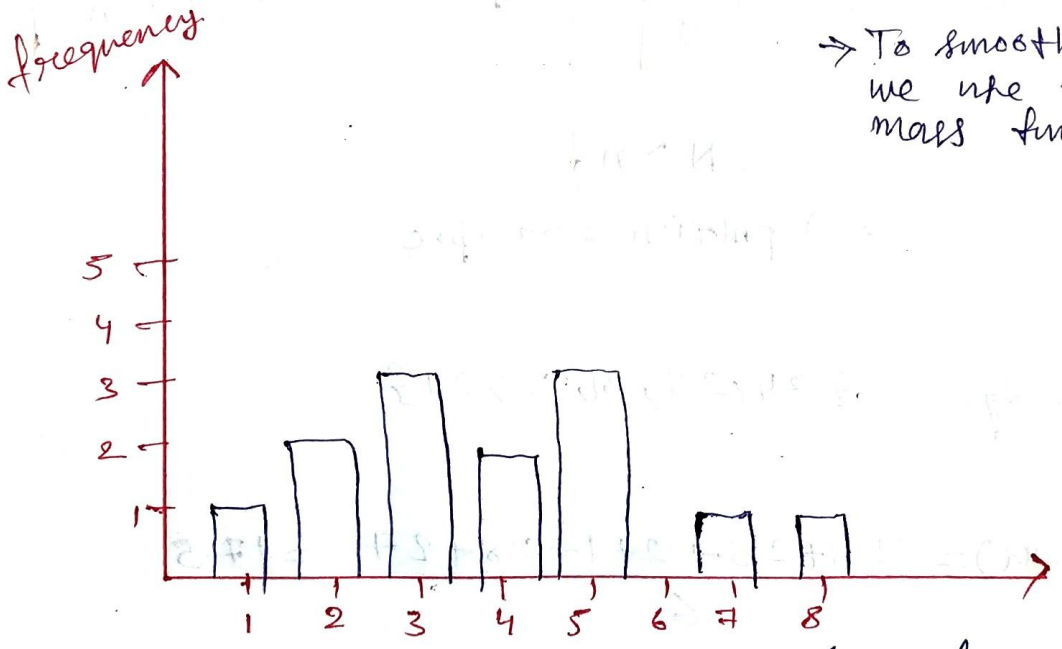
$$\text{bins size} = \frac{95-30}{10} = \frac{65}{10} = 6.5$$



Discrete continuous

Ranks = {1, 2, 3, 4, 5, 6}

Number of bank accounts = [2, 3, 5, 1, 1, 4, 5, 3, 7, 8, 3, 2, 4, 5]



→ To smoothen this, we use probability mass function (PMF).

→ There is no decimal value in discrete continuous.

→ There is gaps between different blocks.

Note: For continuous, use Probability Density Function (PDF)
For Discrete, use Probability Mass Function (PMF)

Measure of Central Tendency

1. Mean
2. Median
3. Mode

- A measure of central tendency is a single value that attempts to describe a set of data identifying the central position.

Mean

$x = \{1, 2, 3, 4, 5\}$

Average / mean $= \frac{[1+2+3+4+5]}{5} = \frac{15}{5} = 3$ (central element in the dataset)

Population (N)

Population mean $(\mu) = \sum_{i=1}^{N} \frac{x_i}{N}$

Sample (n)

Sample mean $(\bar{x}) = \sum_{i=1}^{n} \frac{x_i}{n}$

$N \geq n$

$\Rightarrow$ Population $\geq$ Sample

e.g.:

1. Population Age $= \{24, 23, 21, 28, 27\}$

$N = 6$

Population Mean $(\mu) = \frac{24+23+21+28+27}{6} = 17.5$

Sample Age = {24, 21, 1, 27} – (Random sampling from Population Age dataset)

$n = 4$

Sample Mean $(\bar{x}) = \frac{24+21+1+27}{4} = 13.5$

Relationship b/w μ and $\bar{x}$

$\mu \gtrless \bar{x}$
$\bar{x} \gtrless \mu$
} Both can happen)

$N \geq n$ (Always)

→ There is no fixed relationship b/w μ and $\bar{x}$.

Practical Application (Feature Engineering)

Age	Salary	Family size
–	–	–
–	–	–
–	–	–
NAN	–	–
–	–	–
–	–	–
–	NAN	–
–	–	NAN
–	NAN	–
NAN	–	–

{All are numerical column}

→ We cannot drop the nan values, if we do the loss of information will happen.

→ We can replace the NAN with the mean of that column.

Median

e.g.

Age	Salary
24	45
28	50
29	NAN
NAN	60
31	75
36	80
NAN	NAN
80	200

$$\text{Age_Mean} = \frac{24+28+29+31+36}{5} = 29.6$$

$$\text{Salary_Mean} = \frac{45+50+60+75+80}{5} = 62$$

After the inclusion of 80 and 200,

~~Age_mean =~~

$$\text{Age_mean} = \frac{24+28+29+31+36+80}{6} = 38$$

$$\text{Salary_mean} = \frac{45+50+60+75+80+200}{6} = 85$$

→ Because of outliers 80 and 200, mean value has increased by huge margin.

→ In order to prevent this problem, we use **median**.

e.g. $\{1, 2, 3, 4, 5\}$ $\{1, 2, 3, 4, 5, 100\}$ ← outlier

$\bar{x} = 3$ ⟶ $\bar{x} = \frac{1+2+3+4+5+100}{6}$

$= 19.16$

Steps to find out median:

① Sort the number

② Find the central number
   (1.1) if the number of elements are even, we find the average of central elements.
   (1.2) if the no. of elements are odd, we find the central elements.

e.g { 1, 2, 3, 4, 5, 6, 7, 8, 100, 120 }

① Already sorted

② central element = $\frac{5+6}{2}$ = 5.5. [n = 10]

Mean = ~~25.6~~ 42.66

e.g { 0, 1, 2, 3, 4, 5, 6, 7, 8, 100, 120 }

① Already sorted

② central element = 5 [n = 11]

Mean = 36.57

[Note: If outliers present → Use median
No outliers present → Use mean.]

Mode

~~It~~ is defined as the most frequent occuring elements.

e.g: { 1, 2, 2, [3, 3, 3], 4, 5 }
mode = 3

e.g: { 1, 2, 2, 2, 3, 3, 3, 4, 5 }
Mode = 2, 3

Practical implementation:

→ Useful in categorical variable.

Types of flower:

Lily
Sunflower
Rose
NAN → Replace with Rose
Rose
Sunflower
Rose
NAN → Replace with Rose

Frequency of Rose = 3 (maximum)

Mode = Rose.

[NOTE: If more than 90% of data are NAN, then delete that column. This also depends on the problem statement. Generally, we delete the column if more than 90% values are NAN.]

Measure of Dispersion

① Variance (σ^2)

② Standard Deviation (σ)

Variance (σ^2)

→ Spread of data

Population Variance (σ^2)

$$\sigma^2 = \sum_{i=1}^{N} \frac{(x_i - \mu)^2}{N}$$

Sample Variance (s^2)

$$s^2 = \sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n-1}$$

$x_i - \mu$ → Distance from the mean of every data points

Eg Which dataset has high variance?

(a) {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

↓

Variance

mean = 5.5

(b) {1, 2, 3, 4, 50, 60, 70, 100}

↓

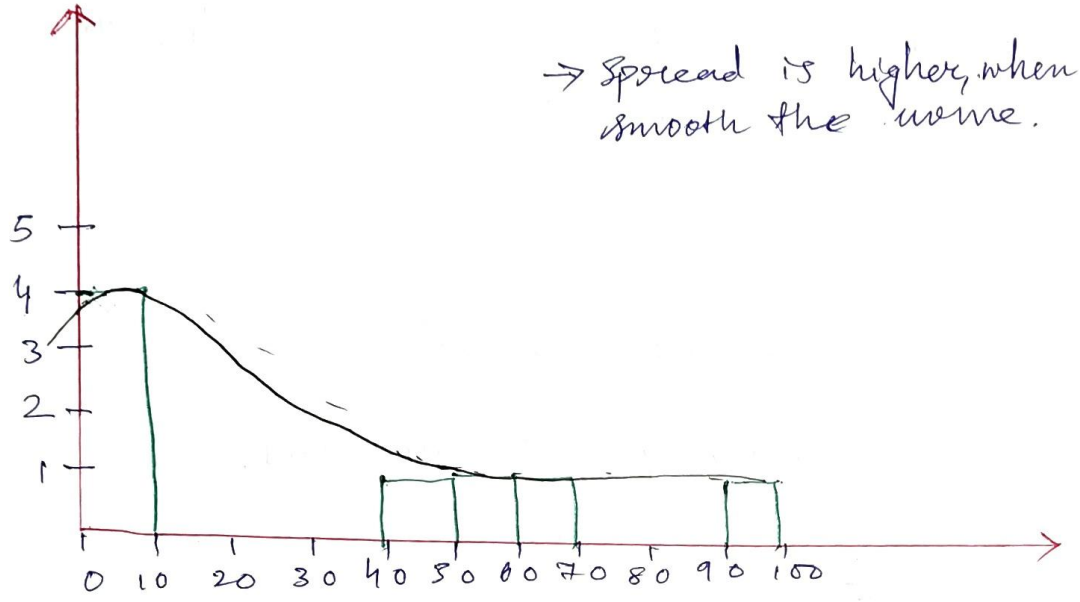
Variance

mean = 36.25.

→ variance will be higher.

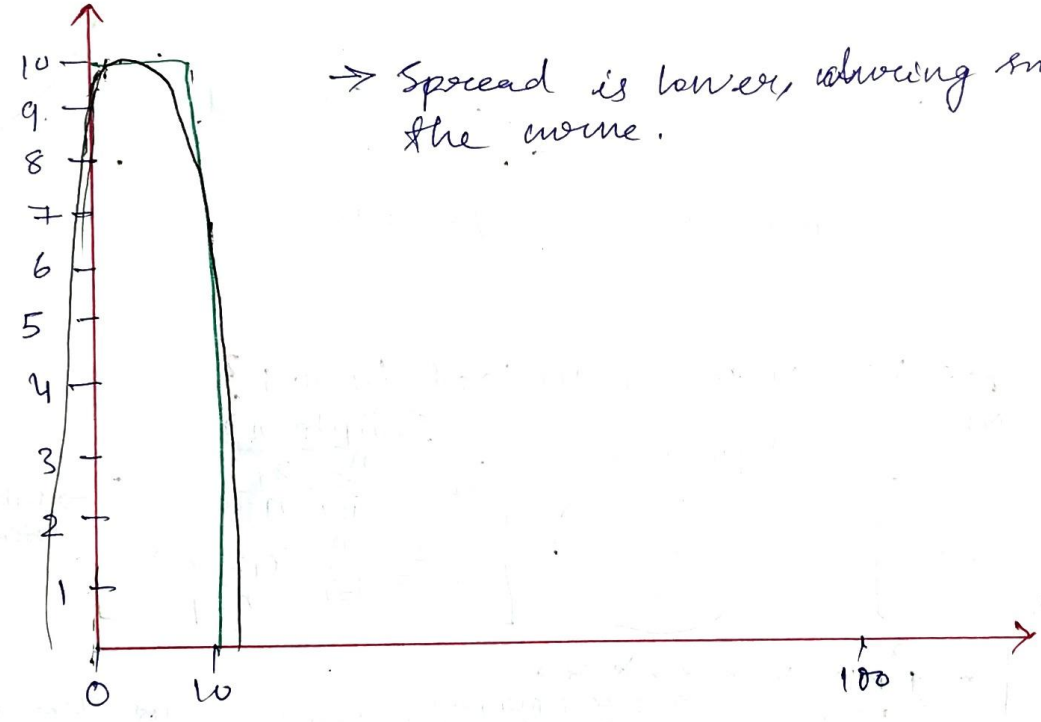
Histogram

(b) Bins size = 10



→ Spread is higher, when we smooth the curve.

(a) Bins size = 1



→ Spread is lower, showing smoothing of the curve.

e.g: (a) $\{1, 2, 3, 4, 5\}$ (b) $\{1, 2, 3, 4, 5, 6, 80\}$

$\mu_a = 3$ $\mu_b = 14.4$

$$\sigma_a^2 = \frac{\{(1-3)^2 + (2-3)^2 + (3-3)^2 + (4-3)^2 + (5-3)^2\}}{5}$$

$$\sigma_a^2 = \frac{10}{5} = 2$$

$$\sigma_b^2 = \{(1-14.4)^2 + (2-14.4)^2 + (3-14.4)^2 + (4-14.4)^2 + (5-14.4)^2 + (6-14.4)^2 + (80-14.4)^2\} / 7$$

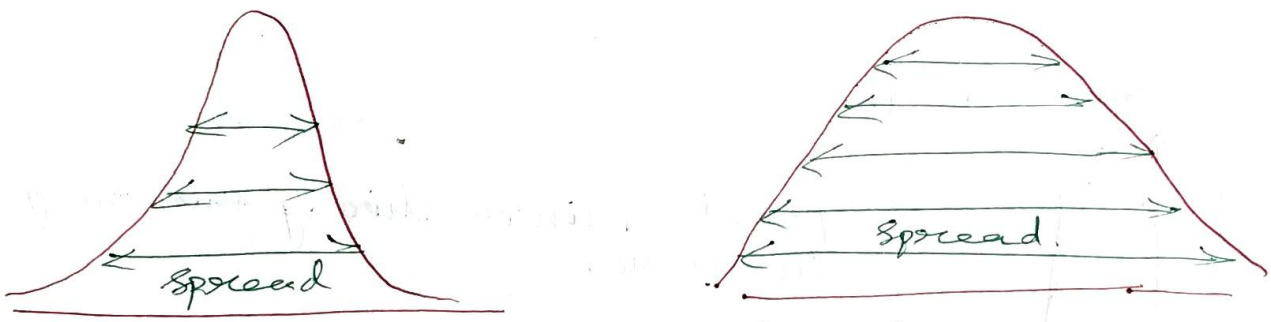
$$= 719.40$$

→ Spread is high in "b" dataset.

→ As variance increase, spread increases.

e.g:

$\sigma^2 < \sigma^2$



Spread < Spread.

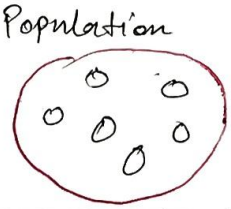
Assignment

Que Why sample variance is divided by n-1?

→ Population (N)

$$\mu = \frac{\sum_{i=1}^{N} x_i}{n}$$

$$\sigma^2 = \frac{\sum_{i=1}^{N} (x_i - \mu_x)^2}{N}$$

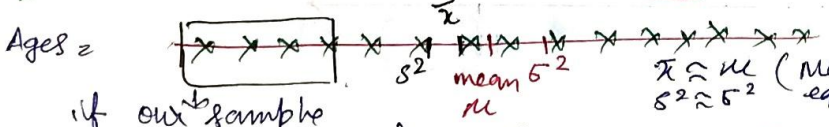


Sample (n)

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$$

→ unbiased estimator



$\bar{x} \approx \mu$ (May be equal), if data set is not skewed).
$s^2 \approx \sigma^2$

if our sample is only 4 points from starting then sample variance and population variance varies a lot.

→ divided by (n-1) is to get sample variance close to population variance.

} This are generally happened in skewed data.

Standard Deviation (σ)

$$\sigma = \sqrt{\sigma^2}$$

→ It tells that, how many standard deviation a data fall from the mean.

eg: $\{1, 2, 3, 4, 5\}$

$\mu = (1+2+3+4+5)/5 = 3$

$$\sigma^2 = \frac{\{(1-3)^2 + (2-3)^2 + (3-3)^2 + (4-3)^2 + (5-3)^2\}}{5}$$

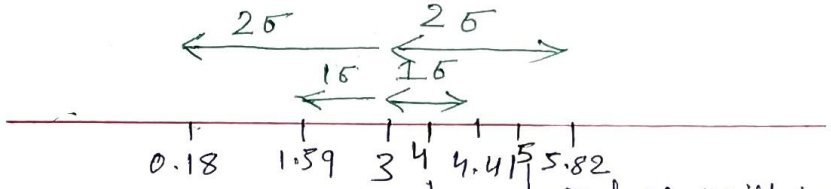
$$= \frac{4+1+0+1+4}{5}$$

$$= 2$$

$$\sigma = \sqrt{\sigma^2} = \sqrt{2} = 1.41$$

Variance = 2

Standard deviation = $\sqrt{2}$ = 1.41



→ fall within 2 S.D to the right from the mean.

fall with in 1st S.D to the right from the mean.

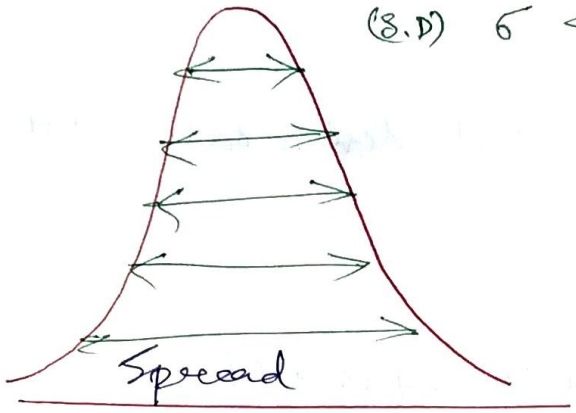
3 → 0 S.D away from the mean.

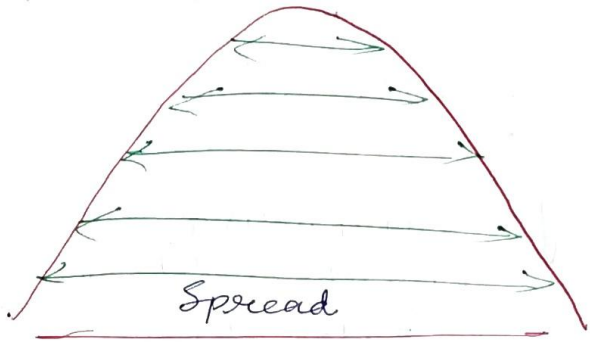
→ By naked eye also, we can tell the spread, but with the help of we can become.

e.g:

$\sigma^2 < \sigma^2$

(S.D) $\sigma < \sigma$ (S.D)





Percentiles and Quartiles

Percentiles

Percentage = {1, 2, 3, 4, 5, 6, 7, 8}

Percentage of even number = $\frac{\text{No. of even numbers}}{\text{Total no. of number}}$

$= \frac{4}{8} = 0.5 = 50\%$

Percentiles: GATE, CAT, FBLTS, JEE, NEET etc

→ A percentile is a value below which a certain percentage of observations lie.

99 percentile: It means the person has got better marks than 99% of people.

eg data = 2, 2, 3, 4, 5, 5, 5, 6, 7, 8, 8, 8, 8, 8, 9, 9, 10, 11, 11, 12

What is the percentile rank of 10?

→ Percentile rank of x = $\frac{\text{No. of value below } x}{\text{Total no. of value}}$

Percentile of 10 = $\frac{16}{20} = 80$ percentile.

→ Rank of 10 is better than 80 percentage of the people.

what is the percentile rank of 8?

→ Percentile rank of 8 = $\frac{9}{20}$ = 0.45 = 45%.

→ Rank of 8 is better than 45% of the people.

what is the percentile rank of 6?

→ Percentile rank of 6 = $\frac{7}{20}$ = 0.35 = 35%.

→ Rank of 6 is better than 35% of the people.

Note: To count no. of data below 6, don't include 6.

e.g. what is the value that exists at 25 percentile?

For ~~odd~~ even → $\text{value} = \frac{\text{Percentile}}{100} \times (n+1)$ → It will give index of the element.

n = 20

{Doubt in n or (n+1)}

$\text{value} = \frac{25}{100} \times (20+1)$ 5th index = 5.25 = 5th index

= 5th index

what is the value that exists at 95 percentile?

→ $\text{value} = \frac{95}{100} \times (20+1)$ 19th index = 19.95 = 19th index = 11

= 19th index

~~For even~~ For ODD → $\text{value} = \frac{\text{Percentile}}{100} \times (n+1)$

~~let's n=~~ let's n = 21

5 number Summary

1. Minimum
2. 1st Quartile (25 percentile) [Q1]
3. Median
4. Third Quartile (75 percentile) [Q2]
5. Maximum

→ Using all these, we can remove outliers. by creating box plot.

index→ 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

e.g. {-15, 1, 2, 2, 2, [3, 3], 3, 4, 5, 5, 5, 6, 6, 6, 6, [7, 8], 8, 9, 27}

n = 21 → average = $\frac{3+3}{2}$ = 3 → average = $\frac{7+8}{2}$ = 7.5

Lower Fence ⟷ Higher Fence

Lower Fence = Q1 − IQR × 1.5
Higher Fence = Q3 + IQR × 1.5

[IQR = Q3 − Q1] ↓ Inter Quartile Range

(doubt n or n+1) → for now, let's take n+1

$$Q1 = \frac{25}{100} \times (n+1)$$

$$= \frac{25}{100} \times (21+1)$$

$= 5.5^{th}$ index

$= 3$ (average of $\frac{3+3}{2}$, because 5.5th index doesn't exist)

$$Q2 = \frac{75}{100} \times (21+1)$$

$= 16.5^{th}$ index

$= 7.5$

$IQR = Q3 - Q1 = 7.5 - 3 = 4.5$

Lower Fence = $3 - 1.5 \times 4.5 = -3.75$

Higher Fence = $7.5 + 1.5 \times 4.5 = 14.25$

→ All the values which are outside lower and higher fence boundaries are treated as outliers.

Here, outliers are -15 and 27.

e.g

5 number summary:

{-15, 1, 2, 2, 2, 3, 3, 3, 4, 5, 5, 5, 6, 6, 6, 6, 7, 8, 8, 9, 27}

→ All the below calculations are without outliers.

① Minimum = 1

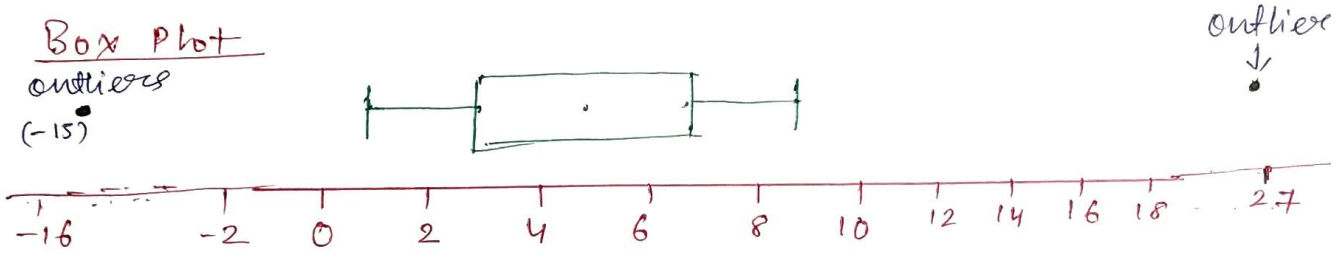
② Q1 = 3

③ Median = 5

④ Q3 = 7.5

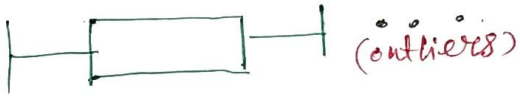
⑤ Maximum = 9

Box Plot



Box Plot (In Python)

→
```
import seaborn as sns
df = sns.load_dataset('tips')
sns.boxplot(df['total_bill'])
```



~~Percentile (In Python)~~